

Three Evolution Mechanisms in Productive Tension: The Integrating Frame for Mutation, Directed Selection, and Action-Feedback Evolution in the Coordination Knowledge Substrate Pattern

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Date: May 7, 2026

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, as a foundational architectural commitment, the integrating frame under which Paper 2's three architecturally distinct evolution mechanisms — mutation, directed selection, and action-feedback evolution — operate in productive tension within one CKS-governed AI Self.

Abstract

Paper 2 commits an AI Self to evolve through three architecturally distinct mechanisms operating concurrently on architecturally separated layers under unified human governance. *Mutation* — instinct evolution — operates on the LLM and substrate-platform infrastructure layer through upstream upgrades the Self does not author. *Directed selection* — DNA evolution — operates on stabilized orchestration content under explicit governance-defined goals. *Action-feedback evolution* closes the loop from accumulated action-layer evidence into governed DNA refinement. Conventional AI evolution frameworks typically commit to one mechanism — model retraining — collapsing what are architecturally distinct dynamics into one operation on one layer. Biology has only mutation; CKS adds directedness alongside undirectedness, with two of its three mechanisms committing to direction under explicit governance-defined goals. This note formalizes the integrating frame as a standalone foundational commitment: it states what the three-mechanism architecture commits to, what the productive-tension framing means and does not mean, where the biological analog is exceeded, what Paper 1 commitments are inherited and in what shape, and what the operational implications and limits are.

1. Why the integrating frame needs to be formalized as a standalone commitment

Paper 2's three evolution mechanisms are easy to misread in two ways, both corrosive to the rest of Paper 2. The first misreading collapses them into one operation parameterized by different inputs — "the system updates itself," sometimes from upstream model upgrades, sometimes from human-authored rule changes, sometimes from accumulated experience. Read this way, the three

mechanisms become implementation details of a single update function, the architectural separation between layers (instinct/reasoning per Paper 2 §4, DNA/action per §5.3) appears incidental rather than load-bearing, and the governance implications collapse with it: one update function admits one governance shape, and the distinct shapes Paper 2 §8 carries become difficult to motivate.

The second misreading treats the three mechanisms as a list rather than an integrated frame. Each is named and operationalized, but the architectural property that *holds them together* — that each contributes what the others cannot, on a layer the others do not touch, under a governance shape the others do not require — is left unstated. Read this way, the next three notes (B1.13 on mutation, B1.14 on directed selection, B1.15 on action-feedback) become independent definitions rather than elaborations of a coordinated architecture.

The remedy is to name the integrating frame explicitly as a foundational commitment in its own right. Three mechanisms operating on three architecturally separated layers under three distinct governance shapes is not coincidence; it is the architecture of evolution Paper 2 commits the Self to, and the rest of Paper 2's evolution machinery rests on it.

2. The architectural commitment, precisely stated

In Paper 2, a CKS-governed AI Self evolves through **three architecturally distinct mechanisms operating concurrently in productive tension across architecturally separated layers under unified human governance applied in distinct shapes per mechanism**. The commitment has six load-bearing components.

(a) Three mechanisms, not one. The mechanisms are *mutation* (instinct evolution), *directed selection* (DNA evolution), and *action-feedback evolution*. They are not variants of one underlying operation; they are architecturally distinct dynamics admitting different actor structures, different temporal profiles, and different governance shapes.

(b) Distinct layers per mechanism. Mutation operates on the instinct/infrastructure layer — the LLM and the substrate-platform infrastructure hosting the substrate — through upstream upgrades the Self does not author. Directed selection operates on the DNA layer of cells, aspects, and Selves — stabilized orchestration and behavior substrate content. Action-feedback evolution operates on the Action-to-DNA pathway, with accumulated action-layer evidence flowing into governed proposals to refine DNA. The layer separation is what Paper 2 §4 (instinct/reasoning) and §5.3 (DNA/action within every cell) make architecturally available.

(c) Distinct dynamics per mechanism. Mutation provides exploratory capacity — undirected change discovers possibilities directed selection would not search for. Directed selection provides directional capacity — purposeful change toward goals defined by human governance. Action-feedback evolution provides empirical refinement — operational experience grounding both the directional change and the integration of new instinct layers.

(d) Distinct governance shapes under unified A1.01 governance. All three inherit Paper 1's human-governed commitment in distinct operational shapes per Paper 2 §8: mutation through verification substrates, routing rules, and the orchestration rules that pin high-stakes decisions to the reasoning layer; directed selection through the standard authority architecture from Paper 1; action-feedback evolution through humans governing the substrates that propose DNA changes from

action evidence and approving the changes — making it human-mediated rather than automatic, which prevents the action layer from silently drifting the DNA over time.

(e) Concurrent operation, not serialized. A Self can absorb an LLM upgrade through mutation governance, accept a DNA refinement through directed selection, and integrate an action-feedback proposal through human-mediated approval all within the same operational window, because the layers are architecturally separated and the governance shapes do not interfere.

(f) Productive tension as architectural property, not incidental result. The three are committed to *together* because the territory they cover collectively is not coverable by any subset. Mutation alone yields undirected variation without trajectory. Directed selection alone yields trajectory without exploration. Action-feedback alone yields experiential refinement without the variation or trajectory experience operates within. Productive tension names what the conjunction produces.

The commitment is *architectural*: it must be a property of the Self's design, not a procedural promise dependent on a particular deployment configuration. A Self in which one of the three mechanisms is architecturally unavailable has collapsed three mechanisms into fewer, and the rest of Paper 2's evolution properties hold only in the residual.

3. What productive tension means and does not mean

The phrase carries specific architectural content that is easy to mis-hear in either of two ways.

Tension does not mean conflict. Productive tension is not adversarial equilibrium between competing forces, not internal opposition resolved by selection pressure, not friction the architecture tolerates as the cost of having multiple mechanisms. The mechanisms operate on architecturally separated layers and admit architecturally distinct governance shapes; they do not compete for the same resource and do not require resolution against one another. *Tension* in Paper 2's sense names compositional integration of distinct mechanisms whose contributions do not collapse into one another.

Tension does not mean automatic balance. The three mechanisms are not self-balancing. Productive tension is not a homeostatic property the architecture maintains without intervention; it is a frame humans configure into operational shape per deployment. A deployment may run one mechanism much harder than another, suppress one in some scopes, or emphasize different mechanisms in different aspect-domains of the same Self. The configuration is governance content, authored and modifiable like any other governance content.

What productive tension *does* mean is that the three mechanisms, when held in conjunction, cover an evolutionary territory none covers alone. Mutation supplies capability the directed process could not have planned for; directed selection supplies trajectory mutation cannot provide; action-feedback supplies operational evidence mutation and directed selection both lack. The conjunction is what Paper 2 §7 commits the Self to, and "productive" names the architectural result of holding it.

4. The biological analog and where CKS exceeds biology

Paper 2's biology vocabulary earns reader comprehension by parallel structure rather than identity. The three-mechanism frame is one of the points where the parallel is bounded explicitly: biology has only mutation, and CKS exceeds biology by adding directedness alongside undirectedness.

Biology's evolutionary mechanism, in the architectural sense Paper 2 uses, is one — natural selection acting on undirected mutation. Extended Evolutionary Synthesis extensions (niche construction, developmental plasticity, inclusive inheritance) bias and bootstrap selection's foundation rather than supply parallel directed-selection mechanisms; selection in biology remains undirected even when developmental plasticity or niche construction shapes the variation it operates on. Biology has no goals because biology has no goal-setter; selection has no direction because there is no authority to direct it.

CKS commits to directedness because CKS has goal-setters — humans, exercising the authority Paper 1 §3.3 commits the architecture to. Two of CKS's three mechanisms are directed: directed selection on the DNA layer toward governance-defined goals, action-feedback evolution toward refinements humans authorize from accumulated operational evidence. Direction here is *external* (humans set the goals) and *persistent* (governance authority is ongoing rather than front-loaded into a configuration that runs autonomously). The third mechanism — mutation — is undirected, parallel to biology, but bounded by governance over how it is integrated rather than left autonomous. The combination does work biology cannot: biology's mutation-plus-selection over deep time achieves adaptive trajectory through accumulated variation, but on evolutionary timescales spanning millions of generations through mechanisms that are not human-mediated and not governable. CKS's human-mediated directedness operates on operational timescales, with substrate edits taking effect immediately under governance, which is what makes evolutionary dynamics architecturally available within the operational lifetime of a deployment.

5. Inherited Paper 1 commitments at the integrating-frame level

The integrating frame inherits Paper 1's foundational commitments without amendment, with the inheritance structurally significant at four points.

A1.01 (human-governed) applies to all three mechanisms in distinct shapes. The same authority architecture from Paper 1 — three rights at all times, governance as authority not labor — applies to each mechanism. What differs across mechanisms is the operational shape governance takes, not the underlying authority architecture.

A1.07 (path retraceability) holds across all three mechanisms. Evolution events at every mechanism are recorded with provenance metadata distinguishing the mechanism, the actor structure (human author, LLM operating under direction, upstream upgrade integration), and the affected substrate content. Every change to substrate content from any of the three mechanisms is traceable to its mechanism and its authorizing actor.

A1.10 (determinism contract) is preserved across mechanism boundaries. Each mechanism has a specification under which its outputs are deterministic given its inputs and the substrate state it operates over: verification substrates with deterministic pass/fail logic for mutation, authority architectures with deterministic decision processes for directed selection, deterministic proposal-and-acceptance machinery for action-feedback.

A2.04 (rule authoring) underwrites two of the three mechanisms directly. The orchestration rules that govern directed selection on DNA, and the rules that govern action-feedback's proposal-and-approval cycle, are authored by humans under the standard authority architecture A2.04 commits to. LLMs may draft rule changes under human direction; humans hold authority over

acceptance.

The integrating frame does not amend Paper 1; it specifies where each Paper 1 commitment binds in the evolutionary dynamics, allowing Paper 2 to be a *composition* with Paper 1 rather than a parallel theory.

6. Operational implications

Five implications follow directly from the integrating frame.

Per-mechanism governance is configurable per deployment. Each mechanism admits its own governance configuration — verification substrates and routing rules for mutation, authority architecture for directed selection, proposal-and-approval machinery for action-feedback. Deployments configure each independently within the unified A1.01 commitment.

Deployments emphasize mechanisms by operational character. A high-stakes regulated deployment may emphasize directed selection with strict mutation containment, with verification substrates running reasoning and instinct decisions in parallel on safety-critical paths. A high-throughput deployment may emphasize mutation's capability benefits with verification routing concentrated where errors are consequential. A learning-from-experience deployment may emphasize action-feedback with strong proposal review.

Mechanisms operate concurrently rather than serially. Multiple mechanisms can affect the same Self in the same operational window without interference because the layers are architecturally separated and the governance shapes do not collide.

Evolution events at each mechanism are recorded distinguishably. Provenance metadata distinguishes which mechanism produced each substrate change, which actor structure authored it, and which governance shape authorized it, enabling audit, replay, and reproducibility across evolutionary dynamics.

The mechanisms apply at the levels where they are appropriate. Mutation operates on the instinct/infrastructure layer that is universal to the Self. Directed selection operates on the DNA layer at any level — cell, aspect, Self. Action-feedback evolution operates on the Action-to-DNA pathway at the level where action accumulates.

7. Limits

Four limits bound the integrating frame and prevent it from carrying weight it does not bear.

The three mechanisms are not optional. Paper 2 commits to all three as architecturally available; a Self that admits only one or two has collapsed the frame, and the rest of Paper 2's evolution dynamics hold only in the residual.

Productive tension does not produce automatic balance. The frame names the conjunction; it does not commit the architecture to balancing the mechanisms autonomously. Humans configure the operational mix and modify it under governance; tension does not self-tune.

The mechanisms do not eliminate evolution risks. Undirected mutation can produce undesired drift if instinct integration is insufficiently governed. Directed selection can produce human errors if rule-authoring is unaudited. Action-feedback can encode pathological experiential patterns if proposal

review is weak. The frame names the architecture; it does not warrant outcomes.

The three mechanisms do not exhaust possible evolutionary dynamics. Additional mechanisms may be authored per deployment requirements without violating the frame; the commitment is to the three Paper 2 specifies as architecturally available, not to a closure claim against further mechanisms.

8. Operational test

A CKS-governed AI Self instantiates the integrating-frame commitment if and only if all of the following are true:

1. The architecture admits mutation as a distinct mechanism operating on the instinct/infrastructure layer through upstream upgrades, governed through verification substrates and routing rules under human authority.
2. The architecture admits directed selection as a distinct mechanism operating on the DNA layer under explicit goals defined by human governance, with selection criteria themselves substrate content under the same authority architecture.
3. The architecture admits action-feedback evolution as a distinct mechanism operating on the Action-to-DNA pathway, with proposal-and-approval machinery under human-mediated authority.
4. The three mechanisms operate concurrently rather than serially, with distinct governance shapes inheriting Paper 1's A1.01 commitment.
5. Evolution events at each mechanism are recorded with provenance distinguishing the mechanism, preserving A1.07 retraceability across evolutionary dynamics.

A Self that fails any of (1)–(5) may evolve in some other architecture's sense, but does not instantiate the three-mechanisms-in-productive-tension frame Paper 2 commits to.

9. Conclusion

The three evolution mechanisms — mutation, directed selection, action-feedback — operating in productive tension across architecturally separated layers under unified human governance applied in distinct shapes are Paper 2's foundational commitment for evolutionary dynamics. The frame is architectural, not procedural. Its content is what makes the three per-mechanism elaborations that follow — B1.13 on mutation, B1.14 on directed selection, B1.15 on action-feedback — coherent as parts of one integrated architecture rather than independent definitions, and what makes the bidirectional-evolution treatment in B1.16 and the structural properties in B1.17–B1.20 land on a stable footing. Naming the frame as a standalone commitment prevents the mechanisms from being read as variants of one underlying update operation, bounds the biology vocabulary at the right architectural depth, and carries the inheritance from Paper 1 across the evolutionary dynamics intact.

Subsequent work that adopts Paper 2's evolution architecture should treat the three mechanisms in productive tension as the integrating frame the per-mechanism content elaborates within. Subsequent work that uses fewer mechanisms, merges them into one, or treats their conjunction as incidental is using a different architecture; the difference should be named.

Source paper

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Three Evolution Mechanisms in Productive Tension: The Integrating Frame for Mutation, Directed Selection, and Action-Feedback Evolution in the Coordination Knowledge Substrate Pattern*. May 7, 2026. ORCID: 0009-0004-8065-3235.